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Software Requirements Specification (SRS)

Project Oracle: AI-Native Trading Architecture

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1. Introduction

1.1 Purpose

This document describes the external and internal requirements for Project Oracle, an AI integration layer for TradingView. It is intended for the engineering team, product owners, and stakeholders.

1.2 Scope

The system will provide a real-time AI interface within the TradingView Advanced Charts Library, featuring generative script coding, vision-based pattern analysis, and contextual sentiment scoring.

2. Overall Description

2.1 Product Perspective

Project Oracle is a modular extension of the Dante Labs platform that interfaces with TradingView via the Universal Data Feed (UDF) and custom JavaScript event listeners.

2.2 User Classes and Characteristics

- **Retail Traders:** Require intuitive NL-to-Code interfaces and visual signal overlays.
- **Quantitative Analysts:** Require API access to AI-verified signal outputs for systematic testing.

3. System Features

[SF-01] Natural Language Pine Scripting (NLPS)

The system shall interpret English language descriptions of trading strategies and compile them into Pine Script v5 code with a >90% syntax accuracy rate.

[SF-02] Computer Vision Pattern Recognition

The system shall analyze the canvas pixel data of current charts to identify classic technical analysis patterns (e.g., Head & Shoulders) and provide a confidence interval overlay.

[SF-03] Real-time Social Sentiment Mapping

The system shall correlate real-time social media volume spikes with price action, displaying heatmaps directly on the candle chart.

4. External Interface Requirements

4.1 User Interfaces

- **TradingView Pane:** A custom React-based side panel embedded within the TradingView library frame.
- **Interactive Overlays:** Direct Canvas manipulation for drawing AI-predicted price trajectories.

4.2 Software Interfaces

- TradingView Lib

Interface	Protocol	Requirement
Market Data	WebSockets	Must support 100ms refresh for Binance/CME.
AI LLM	gRPC / CUDA	Model inference must complete in < 500ms.
JavaScript API	Must support Chart Library v26+.	

5. Non-functional Requirements

5.1 Performance Requirements

- **Latency:** End-to-end signal latency must not exceed 250ms (Ingestion to UI).
- **Concurrency:** System must support 10,000 concurrent websocket connections per cluster node.

5.2 Safety & Security Requirements

- **Strategy Privacy:** User-generated AI strategies must be encrypted at rest (AES-256).
- **Rate Limiting:** LLM requests shall be limited to 60 requests per minute per user to prevent API flooding.

6. Functional Requirement Traceability Matrix

Req ID	Description	Priority	Complexity
FR-01	NL to Pine Script v5 compilation	High	High
FR-02	Historical pattern backtesting via AI	Medium	Medium
FR-03	Multi-asset correlation heatmap	Medium	Low
FR-04	Automated Stop-Loss/Take-Profit AI suggestions	High	Medium